Adify Skills
n8n Automation Mastery

📘 CLASS NOTES

🛡️ Module 6 · Production-Grade Engineering

DAY 20 of 25

# Scaling, Security & Workflow DevOps

"Ek client se 50 clients tak — bina tootey."

⚡ AAJ HUM YEH BANAYENGE

## Nightly Workflow Backup → GitHub

Har raat 2 baje n8n API se saare workflows nikaalta hai aur har ek ko date-wise folder mein GitHub repo mein commit karta hai. Galti se delete ho jaaye to bhi version wapas.

Redis | Queue Mode | GitHub | n8n API

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

# 🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

**💡 Yeh lesson kyun zaroori hai?**

Ek client se 50 clients tak jaane ke liye scale, security aur version control chahiye — warna ek galti sab tod deti hai.

## 🎯 Aaj ke baad aap yeh kar paoge

**01 Queue mode with Redis & workers**

🌍 Order board + kai chef.
⚙️ EXECUTIONS_MODE=queue, Redis, n8n worker containers.

**02 Performance: batching, memory, binary data mode**

🌍 Ek baar mein 10 plate, 1000 nahi.
⚙️ Split in Batches, binary data mode = filesystem.

**03 Security: credentials, RBAC, secrets, webhook auth**

🌍 Ghar ki chaabi sabko nahi dete.
⚙️ Credentials store, user roles, webhook header/basic auth.

**04 Version control: export JSON, Git, environments**

🌍 Recipe book ki photocopy.
⚙️ n8n API → workflows export JSON → GitHub commit.

**05 Testing workflows & staging vs production**

🌍 Pehle ghar par try, phir party mein.
⚙️ Staging instance → test data → phir production.

**06 Monitoring: execution logs, uptime, metrics**

🌍 Dukaan ka CCTV.
⚙️ Execution logs, uptime ping, failed executions count alert.

## ⏱️ Class ka plan (2 ghante)

| 10m | 35m | 45m | 20m | 10m |
|---|---|---|---|---|

`0:00-0:10` **Recap**

🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

`0:10-0:45` **Concept: Scale Architecture**

🪄 *Ek chef 100 log ka khana nahi bana sakta. 5 chef aur ek order board chahiye.*

`0:45-1:30` **Live Build**

Workflow backup to GitHub (auto, nightly)

🪄 *Aaj ki kahani yaad hai — 'Restaurant ka bada kitchen'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

`1:30-1:50` **Quiz + Audit**

Module 6 quiz + security audit checklist

🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

`1:50-2:00` **Q&A + Homework**

🪄 *Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

# 🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

### Queue mode

Main instance + Redis + multiple workers — parallel executions.

🪄 **Analogy:** Restaurant: ek order-taker, kai chefs

CONCEPT 02

### Binary data mode

Files memory ki jagah disk/S3 par — RAM bachti hai.

🪄 **Analogy:** Saamaan godown mein rakho, kamre mein nahi

CONCEPT 03

### Source control

Workflows JSON ko Git mein — history + rollback.

🪄 **Analogy:** Google Docs version history

CONCEPT 04

### Least privilege

Har credential ko sirf zaroori permission.

🪄 **Analogy:** Guest ko sirf drawing room ki chaabi

### 🔑 Aaj ka main idea

Scale = zyada load sambhalna (queue mode, workers, batching). Security = credentials, access aur webhooks ko surakshit rakhna. DevOps = workflows ka version, testing aur monitoring.

- Redis + 3 workers se 10× throughput
- Webhook par header auth
- Har raat workflows GitHub par backup

## 📖 Ek kahani se samjho: Restaurant ka bada kitchen

1 🧾 **Bahut saare orders aaye** = Executions queue

2 📋 **Order board par lage** = Redis queue

3 👨‍🍳 **5 chef saath kaam karein** = Workers

4 🔐 **Kitchen mein sirf staff** = RBAC + credentials

5 📸 **Har recipe ki copy** = Git version control

🪄 Ek line mein: “Ek chef 100 log ka khana nahi bana sakta. 5 chef aur ek order board chahiye.”

# 🗺️ Aaj ka workflow: Nightly Workflow Backup → GitHub

Data left se right chalta hai. Har dabba ek n8n node hai.



**💰 Isse kitna kama sakte ho?**

Agencies with many workflows (backup + DevOps) — aam taur par **₹5,000 – ₹15,000 add-on**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

## 🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Schedule Trigger**
   Roz raat 2 baje
2. 🌐 **n8n (API node)**
   Get all workflows
3. 🔁 **Loop Over Items (Split in Batches)**
   Har workflow ek-ek karke
4. 👨‍💻 **Code**
   JSON pretty-print, file name = workflow-name.json
5. 🐙 **GitHub**
   Create/update file in backup repo
6. ✈️ **Telegram**
   "✅ 37 workflows backed up"

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

**🔑 Setup (import ke baad)**

- ☐ Connect: **n8n API key (Settings → n8n API)**
- ☐ Connect: **GitHub (personal access token)**
- ☐ GitHub par private repo "n8n-backups" banao
- ☐ n8n Settings → n8n API → API key banao; credential mein Base URL = https://your-n8n-domain/api/v1
- ☐ GitHub token mein "Contents: read & write" permission do

**✅ Test kaise karein**

1. "Test workflow" → GitHub repo mein backups/<aaj ki date>/ folder check karo

## ⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

**😩 Pehle (manual)**
Ek server, ek line mein sab — slow

→

**⚡ Ab (n8n)**
Queue mode: workers parallel

⏱ 1000 webhooks/minute bhi bina atke.

## ⌨️ Aaj ka key snippet

**Copy karke n8n mein paste karo** 📋 Select karke copy karo

```
EXECUTIONS_MODE=queue
QUEUE_BULL_REDIS_HOST=redis
N8N_DEFAULT_BINARY_DATA_MODE=filesystem
```

## 📦 Sample data — input se output tak

**📥 Input** 📋 Select karke copy karo

```
{
  "workflows": 37
}
```

**📤 Output** 📋 Select karke copy karo

```
{
  "committed": 37,
  "repo": "yourbrand/n8n-backup"
}
```

# 💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

**💬 Build it with ChatGPT / Claude** 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Har raat 2 baje n8n API se saare workflows nikaalta hai aur har ek ko date-wise folder mein GitHub repo mein commit karta hai. Galti se delete ho jaaye to bhi version wapas.

Flow: Schedule (2:00 AM) → n8n API (Saare workflows) → Code (File path + JSON) → GitHub (Commit file)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

**💬 Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Nightly Workflow Backup → GitHub" (nodes: Every Night 2 AM, Get All Workflows, Prepare Files, Commit to GitHub).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

## 📥 Workflow ko n8n mein kaise laayein

1. **QR scan karo** ya link kholo — workflow ki JSON file khulegi.
2. Poora text copy karo ( `Ctrl` + `A` , `Ctrl` + `C` ).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

**⚡ Workflow JSON**
4 nodes · ready to import
`adifyskills.com/workflows/day-20-nightly-github-backup.json`

**🌐 Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/20`

# Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

| Agar yeh ho raha hai… | To yeh karo |
|---|---|
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "n8n API key (Settings → n8n API)" connect karo. |
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "GitHub (personal access token)" connect karo. |
| **Time par apne aap nahi chala?** | "Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo. |

### ⚠️ Yeh galtiyan sabse common hain

- ! Production mein direct edit — staging nahi
- ! API keys workflow JSON mein hardcoded (Git par leak)
- ! Execution data kabhi prune na karna — DB full

### ⭐ Pro tips — time bachane wali tricks

- ★ EXECUTIONS_DATA_PRUNE=true + max age set karo
- ★ Credentials kabhi export mat karo Git mein
- ★ Uptime monitoring (UptimeRobot) on /healthz

**Phir bhi atke? Koi baat nahi 🙂**
Screenshot + error message ke saath WhatsApp par poochho →

# 🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

### 🛠 Live lab

Nightly workflow: saare workflows export karke GitHub repo mein commit karna.

### 📝 Homework

Apne instance ka security audit checklist ke hisaab se karo aur report submit karo.

- [ ] Homework ho gaya

**BEGINNER**

Workflow manually export/import

- [ ]

**INTERMEDIATE**

Nightly GitHub backup

- [ ]

**ADVANCED**

Staging → production promotion flow

- [ ]

### 🧩 Aaj ka challenge — pehle khud try karo

**🎯 Backup ke baad Telegram par "✅ 12 workflows backed up" message aaye.**

**💡 Hint 1**

Commit ke baad saare items ko gino.

**💡 Hint 2**

Aggregate ya Code node: return [{json: {count: $input.all().length}}]

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

# 🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

**Q1. Parallel executions ke liye mode?**

- (A) Queue mode
- (B) Manual mode
- (C) Test mode
- (D) Pin mode

**Q2. Queue mode ko kya chahiye?**

- (A) SMTP
- (B) Redis
- (C) MongoDB
- (D) FTP

**Q3. Workflows ki history ke liye?**

- (A) Email
- (B) Pen drive
- (C) Git / source control
- (D) Screenshots

**Q4. DB full hone se bachne ke liye?**

- (A) Bada font
- (B) More nodes
- (C) Kuch nahi
- (D) Execution pruning

## 📖 Aaj ke naye shabd

**Workflow**
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

**Node**
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

**Trigger**
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

**Webhook**
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

**API**
Do apps ke baat karne ka tareeka. Ek app sawaal bhejta hai, doosra jawab (data) deta hai.

**JSON**
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

**Credential**
Kisi app ka login/API key jo n8n safe tareeke se yaad rakhta hai.

**Execution**
Workflow ka ek baar chalna. Har run ka record Executions tab mein dikhta hai.

## ✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 21

**Project Sprint 1: AI Customer Support Desk**

"24×7 support jo kabhi chhutti nahi leta."

☐ **Day 20 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

# 🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

**Q1 → A**
Queue mode

**Q2 → B**
Redis

**Q3 → C**
Git / source control

**Q4 → D**
Execution pruning

### ✅ Challenge ka solution

Commit to GitHub → Code: return [{json: {count: $input.all().length}}] → Telegram "✅ {{ $json.count }} workflows backed up".

## 📚 Official docs

- Queue mode ↗
- Security ↗
- Source control ↗
- n8n public API ↗

## ▶ Videos

- n8n queue mode workers redis ↗
- n8n backup workflows github ↗

### 💬 Doubt poochho

WhatsApp par seedha message

`wa.me/919000000000?text=Day%2020%20(Scaling%2C%20Security%20%26%20Workflow%20DevOps)%20mein%20doubt%20hai%3A%20`

### 🌐 Day 20 lesson page

Quiz, copy buttons, videos

`adifyskills.com/day/20`